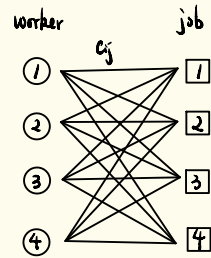


The Assignment Problem

• Problem description

Given n jobs and n workers, we would like to assign each job to one worker, and each worker to one job. If job j is assigned to worker i , the cost of the assignment is C_{ij} .

Our goal is to minimize the total cost of the assignments.



$C = (C_{ij})$ is called the cost matrix of an assignment problem.

• Modeling

Let $x_{ij} = \begin{cases} 1, & \text{if job } j \text{ is assigned to worker } i \\ 0, & \text{if job } j \text{ is NOT assigned to worker } i. \end{cases}$

The assignment problem can be formulated as

$$\min z = \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij}$$

$$\text{s.t. } \sum_{i=1}^n x_{ij} = 1, \quad j=1, \dots, n. \quad (1)$$

$$\sum_{j=1}^n x_{ij} = 1, \quad i=1, \dots, n. \quad (2)$$

$$x_{ij} \in \{0, 1\}, \quad i=1, \dots, n, \quad j=1, \dots, n. \quad (3)$$

Note that the binary constraints (3) can be equivalently replaced with

$$x_{ij} \geq 0, \quad x_{ij} \text{ integer}, \quad i=1, \dots, n, \quad j=1, \dots, n \quad (4)$$

because of (1) and (2). Due to unimodularity, the integer restrictions can be dropped from (4).

That is, the assignment problem can be reformulated as

$$\min z = \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij}$$

$$\text{s.t. } \sum_{i=1}^n x_{ij} = 1, \quad j=1, \dots, n.$$

$$\sum_{j=1}^n x_{ij} = 1, \quad i=1, \dots, n.$$

$$x_{ij} \geq 0, \quad i=1, \dots, n, \quad j=1, \dots, n.$$

Note: The assignment problem is a special transportation problem, where all the supplies and demands are ones.

• Degeneracy

Since the assignment problem is a special transportation problem, we know that it has n^2 variables and $2n-1$ linearly independent constraints. Therefore, a BFS of the assignment problem has $2n-1$ basic variables.

On the other hand, for each assignments, exactly n of the n^2 variables are nonzero. Therefore, every BFS is degenerate. If we apply the simplex method or the transportation algorithm to an assignment problem, many iterations are going to be spent on changing the basis matrix without changing the BFS.

• Assignment matrix

jobs

workers

C_{11}	C_{12}	...	C_{1n}
C_{21}	C_{22}	...	C_{2n}
$\vdots$	$\vdots$	$\ddots$	$\vdots$
C_{n1}	C_{n2}	...	C_{nn}

eg.:

2	10	3	17
5	3	9	10
8	2	5	14
3	5	10	16

• Reduced matrix

Thm: Let C be the cost matrix for an $n \times n$ assignment problem, and let C' be the matrix formed by adding α to each entry in a row/column of C . Then any optimal solution of the assignment problem defined by C is an optimal solution of the assignment problem defined by C' , and vice versa.

e.g.:

$$C = \begin{array}{|c|c|c|} \hline 5 & 7 & 10 \\ \hline 4 & 9 & 6 \\ \hline 8 & 6 & 7 \\ \hline \end{array} \quad \rightarrow \quad C' = \begin{array}{|c|c|c|} \hline 0 & 2 & 5 \\ \hline 4 & 9 & 6 \\ \hline 10 & 8 & 9 \\ \hline \end{array}$$

Pf: Suppose that a constant α is added to the r th row of C to obtain C' .

The objective function for the new problem with cost matrix C' is

$$\begin{aligned} z' &= \sum_{i=1}^n \sum_{j=1}^n C'_{ij} x_{ij} = \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij} + \sum_{j=1}^n C'_{rj} x_{rj} \\ &= \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij} + \sum_{j=1}^n (C_{rj} + \alpha) x_{rj} \\ &= \sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij} + \sum_{j=1}^n \alpha x_{rj} \\ &= z + \alpha. \end{aligned}$$

A constant change in the objective function does not change the optimal solution set.

A similar argument can be made if a constant is added to a column of C .

Def: A reduced matrix has at least one zero in each row and column.

e.g.:

0	8	0	8
2	0	5	0
6	0	2	5
0	2	6	6

To form a reduced matrix C' from C , we can subtract the smallest entry in each row from that row and then subtract the smallest entry in each column from that column.

eg.:

5	7	10
4	9	6
8	6	7

 $\rightarrow$

0	2	5
0	5	2
2	0	1

 $\rightarrow$

0	2	4
0	5	1
2	0	0

Reduced matrices are not unique. For example, we can first subtract the smallest entry in each column from that column and then subtract the smallest entry in each row from that row.

eg.:

5	7	10
4	9	6
8	6	7

 $\rightarrow$

1	1	4
0	3	0
4	0	1

 $\rightarrow$

0	0	3
0	3	0
4	0	1

• Identify assignments

For a reduced matrix, we can find zero-cost assignments by following the procedure below.

- Make an assignment in a zero cell in any row that has only one zero in it. This assignment is noted by placing a box around the zero. Cross out all other zeros in the column containing the box.
- Make an assignment in a zero cell in any column that has only one zero in it by placing a box around the zero. Cross out all other zeros in the row containing the box.
- Repeat a) and b) until no further assignments are possible. If some zeros remain unboxed and uncrossed, but no row/column contains a single uncrossed zero, then make an assignment in an arbitrary zero cell and cross out all other zeros in the row and column containing the assigned (boxed) cell.

eg.:

5	8	0	8
2	9	5	0
6	0	2	5
0	2	6	6

1	1	2	0
3	0	5	4
5	3	3	6
0	1	4	2

0	1	3	2
0	0	5	3
4	0	5	0
6	7	0	1

- Modify the reduced matrix

If n assignments can be made in the previous step, then we have an optimal solution because they are the least-cost (zero-cost) assignments. If only $k < n$ assignments can be made, we need to modify the reduced matrix by covering all zeros with lines as follows:

- Count the number of uncovered zeros in each row and column.
- Draw a line through the row or column with the most uncovered zeros.
- Repeat a) & b) until all zeros are covered.
- Let C_0 be the value of the smallest uncovered element. Subtract C_0 from all uncovered elements. Add C_0 to all elements that are covered by two lines.
(Equivalently, subtract C_0 from all uncovered rows and add C_0 to all covered columns.)

e.g.:

0	2	4
0	5	1
2	0	0

→

0	2	4
0	5	1
3	0	0

→

0	1	3
0	4	0
3	0	0

- The Hungarian algorithm

Step 0. Begin with an assignment matrix.

Step 1. Form a reduced matrix

Step 2. Identify the assignments. If n assignments are possible, STOP. Otherwise, go to Step 3.

Step 3. Modify the reduced matrix.

Return to Step 2.

e.g.:

10	12	11	14
7	13	12	16
12	19	20	18
8	18	12	17

→

0	2	1	4
0	6	5	9
0	7	8	6
0	10	4	9

→

0	0	0	0
0	4	4	5
0	5	7	2
0	8	3	5

←

0	0	0	0
0	4	4	5
0	5	7	2
0	8	3	5

→

2	0	0	0
0	2	2	3
0	3	5	0
0	6	1	3

→

2	0	0	0
0	2	2	3
0	3	5	0
0	6	1	3

←

3	0	0	0
0	1	1	2
1	3	5	0
0	5	0	2

optimal solution:

$$x_{12} = x_{21} = x_{34} = x_{43} = 1$$

all other variables are 0.

optimal value:

$$z = 12 + 7 + 18 + 12 = 49$$